StoryCast: Technical Documentation

How the site is built, how the Sass is organised, and how each accessibility requirement is met.

1. Overview

StoryCast is a static three-page site: index.html, about.html and story pages in story/. No framework is used. Styling is written in Sass and compiled to css/main.css. One small script (js/main.js) enhances transcripts; every page works without it.

2. Project structure

```
storycast/
  index.html              Home page
  about.html              About / Access page
  story/                  Story detail pages (one template, three stories)
  sass/                   Source styles (main.scss + partials)
    abstracts/            _colors, _typography, _spacing, _breakpoints
    base/                 _layers, _reset, _base
    layout/               _layout (container, page shell, header, footer)
    components/           skip-link, site-nav, button, hero, media-card,
                          media-player, transcript, feature, utilities
    pages/                _story, _about
  css/main.css            Compiled output (committed so the site runs without a build)
  js/main.js              Transcript seek + active-line highlight
  assets/                 audio/, video/, captions/ (.vtt), transcripts/ (.txt), images/
  docs/                   research.pdf, documentation.pdf, img/ (figures)
```

3. Semantic HTML

Element	Use
header, nav, main, footer	Page landmarks on every page; nav has aria-label="Main"
article	Each story page's content; each story card on the home page

Element	Use
section + aria-labelledby	Hero, featured stories, transcript, each About section
aside	Related stories
figure / figcaption	Media player with a description of the media
video / audio + track	Native controls, kind="captions", default so captions load on
ol	Transcript cues and steps; breadcrumb

Each page has exactly one h1 and headings never skip a level.

4. Sass architecture

- **Partials and tokens.** abstracts/ holds all design tokens. Components use semantic aliases such as $text and $surface rather than raw colours. Forwarded through abstracts/_index.scss and consumed with @use … as *.
- **Cascade layers.** @layer reset, base, layout, components, pages, utilities; is declared first in base/_layers.scss. Each partial wraps its rules in its layer, so order of the layers, not selector specificity, decides conflicts.
- **BEM naming.** Block, element and modifier classes such as .media-card__title and .story-list--featured. Selectors stay flat and low in specificity.
- **DRY.** A from($bp) mixin wraps min-width media queries; spacing, type and colour come from tokens only.

5. Responsive design

Feature	Implementation
Page shell	Grid with grid-template-rows: auto 1fr auto keeps the footer at the bottom
Card and feature grids	repeat(auto-fit, minmax(min(100%, 18rem), 1fr)): columns adapt without media queries
Story layout	Single column; main + sidebar at 64rem
Header, nav, badges	Flexbox with wrapping

Feature	Implementation
Fluid type	clamp() scale in _typography.scss
Container query	.story-list__item is container-type: inline-size (name story-card). @container story-card (min-width: 30rem) turns the card into a thumbnail-beside-text layout; at 44rem the title grows. The featured home card spans the full row, so it uses the horizontal layout; the same component stacks in the sidebar and on phones.

6. Accessibility checklist (WCAG 2.1 AA)

Requirement	How it is met
1.1.1 Non-text content	Meaningful images have alt text; decorative thumbnails and the hero pattern use alt="" / aria-hidden
1.2.1 / 1.2.2 Captions and alternatives	WebVTT captions for every audio/video, plus an on-page transcript and a downloadable .txt
1.3.1 Info and relationships	Semantic landmarks, lists, headings, figure/figcaption
1.4.1 Use of colour	Current page uses underline + aria-current; active transcript line uses a bar and bold weight; badges contain text
1.4.3 Contrast	All pairs in section 7 are above 4.5:1; UI borders above 3:1
1.4.4 / 1.4.10 Resize and reflow	rem units, fluid layout, no horizontal scroll at 320px
2.1.1 Keyboard	Native controls; transcript timestamps are real buttons
2.4.1 Bypass blocks	Skip link as first focusable element
2.4.2 Page titled	Unique <title> per page
2.4.7 Focus visible	3px dark outline plus a 6px amber halo on all interactive elements; light variant on dark surfaces
2.3.3 / motion	prefers-reduced-motion removes smooth scroll and transitions
3.1.1 Language	<html lang="en">
4.1.2 Name, role, value	Timestamp buttons have aria-label; regions use

Requirement	How it is met
	aria-labelledby; media has aria-describedby

ARIA used only where HTML falls short: aria-current (page and active cue), aria-label / aria-labelledby / aria-describedby, aria-hidden for decoration, and role="list" on lists that have list styling removed (so Safari/VoiceOver still announce them).

7. Colour contrast results

Foreground	Background	Ratio	Result
Ink #14213d	Cream #fdf8f0	15.1:1	Pass (AAA)
Muted #4a5568	Cream #fdf8f0	7.1:1	Pass (AAA)
Muted #4a5568	Sand #f3e8d6	6.2:1	Pass (AA)
Accent #a33a16	Cream #fdf8f0	6.3:1	Pass (AA)
Accent #a33a16	Sand #f3e8d6	5.5:1	Pass (AA)
White	Accent #a33a16	6.6:1	Pass (AA)
White	Ink #14213d	16.0:1	Pass (AAA)
Teal #0f5c63	Cream #fdf8f0	7.3:1	Pass (AAA)
Border #6b7280	Cream #fdf8f0	4.6:1	Pass (UI, 3:1)
Focus amber #ffbf47	Ink #14213d	9.7:1	Pass (UI, 3:1)

8. JavaScript

js/main.js (about 30 lines) finds the story media and the transcript cues. Clicking a timestamp seeks and plays; on timeupdate the cue being spoken gets aria-current="true". Without JavaScript the transcript remains fully readable.

9. Media

The demo narration is generated with text-to-speech from the same sentences used for the WebVTT captions and the transcripts, so the three always match. Real recordings can replace

the files in assets/: keep the file names, re-time the .vtt, and update the transcript text.

10. Testing performed

Check	Result
Sass compiles with no warnings	Pass
All internal links and asset paths resolve	Pass (scripted check)
Layouts at 390px, 768px and 1280px	Pass (headless Chrome screenshots); container query verified on the featured card
Contrast ratios computed from tokens	All pass (section 7)
Keyboard walkthrough	Recommended to repeat manually in your browser; steps are in the README

11. Running locally

```
npm install             # installs Sass
npm run build           # compiles sass/main.scss to css/main.css
npm run watch           # rebuilds on change
python3 -m http.server 8000   # then open http://localhost:8000
```